LAB PRACTICE · BEFORE P3

MPI: one program, four processes, one result

MPI

Message Passing Interface. A standard C library to send messages between processes. Open MPI implements it.

SPMD

Single Program, Multiple Data. The same program runs as N processes. Each one reads its rank, 0 to N−1, and does its part.

Install

`sudo apt install build-essential libopenmpi-dev`

Already on the lab machines.

The goal

4 processes. Process k computes k^2. Process 0 adds the four: 0 + 1 + 4 + 9 = **14**.

hello.c

```
#include <stdio.h>
#include <mpi.h>
int main(int argc, char **argv) {
  int rank, size, mine, total;
  MPI_Init(&argc, &argv);                      // start MPI
  MPI_Comm_rank(MPI_COMM_WORLD, &rank);        // which copy am I
  MPI_Comm_size(MPI_COMM_WORLD, &size);        // how many copies
  mine = rank * rank;                          // my part of the work
  printf("Hello from %d of %d: %d\n",
         rank, size, mine);
  MPI_Reduce(&mine, &total, 1, MPI_INT,        // add every mine
             MPI_SUM, 0, MPI_COMM_WORLD);      // into process 0
  if (rank == 0) printf("Sum: %d\n", total);
  MPI_Finalize();                              // stop MPI
  return 0;
}
```

```
$ mpicc -Wall -o hello hello.c          # gcc + MPI library
$ mpirun -np 4 ./hello                  # start 4 copies
Hello from 0 of 4: 0
Hello from 3 of 4: 9
Hello from 2 of 4: 4
Hello from 1 of 4: 1
Sum: 14          the 4 Hello lines change order in every run;
                 the sum is always 14
```